

N-001 ANALYTIC-BRANCH manifest certification (operator-authorized)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

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1. Purpose and scope

Under the operator directive of 2026-06-23, the canonical N-001 kernel config is certified against A1-ANALYTIC-BRANCH-MANIFEST, promoting A1-PRODUCTION-KERNEL-MANIFEST from T1 OPEN to T6 PROVED CONDITIONAL and licensing A2/A3 to cite the N-001 config as their certified numerical implementation. Reproducible certifier: [codes/foundations/a1_n001_certify.py](#); certificate: [claims/A1-PRODUCTION-KERNEL-MANIFEST/runs/n001_analytic_branch_certificate.json](#).

2. The certified config and verdict

Certificate (A1-N001-ANALYTIC-BRANCH-MANIFEST, CERTIFIED). The config (`r_zero = 0.2190336`, `mu2_shell = 0.005`, `q0 = 0.6801748`, $Z = -0.9252754$, $Y = 1$) satisfies

$$\delta_Z = 0, \quad \delta_m = 4.3 \times 10^{-18}, \quad \delta_r = 0 \leq \varepsilon_Z = \varepsilon_m = \varepsilon_r = 10^{-9}, \quad \text{mu2_shell} = 0.005 > 0, \quad (1)$$

with $Y > 0$, $Z < 0$. The backend module `Math424_AddA_reading_uniqueness` (sha256 `e9c7d8d8..`) yields $D_{\text{code}}(k) = \text{MU2_CANON} + C(k^2 - Q_0^2)^2$ matching the stored polynomial to 4.4×10^{-16} . Hence the N-001 config is the certified A1-ANALYTIC-BRANCH-MANIFEST instance: the numerical implementation of A2 (well-posedness) and A3 (renormalisation) is no longer a mock.

3. Conditional hypotheses

The certification is conditional on two named, separately-tracked hypotheses: A1-N001-TOLERANCE-RATIFIED (operator-ratified $\varepsilon_Z = \varepsilon_m = \varepsilon_r = 10^{-9}$) and A1-BACKEND-HASH-PIN (backend sha256 `e9c7d8d8..`; a backend-source change mismatches the pin and REFUSES re-use until re-certification). These place the result at T6 PROVED CONDITIONAL (not T7).

4. Devil’s-advocate (CLAUDE.md tier-change self-test)

(α) “Circular — the config was derived from the backend, so it trivially passes.” DISMISSED — the verifier reads the config AS-IS and independently re-evaluates D_{code} from the imported module; the corrupted- q_0 test ($9 \times 10^{-2} \gg \text{tol}$) proves the check is load-bearing. The certificate asserts exactly that the stored config faithfully encodes the backend, which is the required property. (β) “Hash-pin is brittle.” VALID-with-mitigation — brittleness is the intended provenance guarantee; the hypothesis A1-BACKEND-HASH-PIN documents that a backend edit forces re-certification (a feature). (γ) “T6 over-claims a numerical config check.” DISMISSED — the gate schema is the analytic identity (A1-KERNEL-IDENTITY, T6) plus positivity (A1-SCALAR-ANALYTIC-BRANCH, T6), and config satisfaction is

executed (9/9) under two explicit named hypotheses; it is correctly T6 PROVED CONDITIONAL, not T7 (textbook-unconditional). Quantitative sanity check: $\delta_m = 4.3 \times 10^{-18} \ll \varepsilon_m = 10^{-9}$ (9-order margin); semantic residual 4.4×10^{-16} .

5. Result footer

Result ID:	A1-N001-ANALYTIC-BRANCH-MANIFEST (CERTIFIED)
Precise statement:	The canonical N-001 config (r_zero=0.2190336, mu2_shell=0.005, q0=0.6801748, Z=-0.9252754, Y=1) satisfies A1-ANALYTIC-BRANCH-MANIFEST: $Y>0, Z<0, \delta_{\{Z,m,r\}} \leq 1e-9, \mu2_shell>0$; backend D_code(k) matches the stored polynomial to $4.4e-16$. It is the certified A2/A3 numerical implementation.
Scope:	ANALYTIC-BRANCH; operator-authorized 2026-06-23.
Dependencies:	A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Hypotheses:	A1-N001-TOLERANCE-RATIFIED; A1-BACKEND-HASH-PIN.
Evidence grade:	ANALYTIC + EXECUTED (certifier CERTIFIED; verifier 9/9).
Reproduction command:	python codes/foundations/a1_n001_certify.py
Expected output:	CERTIFIED; ANALYTIC-BRANCH=True; exit 0.
Tier:	T6 PROVED CONDITIONAL (operator-authorized).
Falsifier:	a backend-source change (hash mismatch) or a config edit breaking any delta gate REFUSES certification.
No-overclaim:	certified for the ANALYTIC-BRANCH scope only; not a TOE claim; condensate-branch ($\mu2_shell<0$) configs excluded.
Next required action:	A2/A3 cite the N-001 config as the certified numerical implementation; mainline production-Hessian may build on it.